

Particle PhysicsNotes

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Lecture 7

- QFT $\rightarrow$ Calculation tool, provides us with tools to calculate decay rates and scattering cross-section
- This needs an input $\rightarrow$ Lagrangian
- To write lagrangian you need constraints
- Symmetries provide these constraints

Lorentz Group

- $J_i \rightarrow$ Rotation Generators, $K_i \rightarrow$ Boost Generators
- Lorentz Group is $SO(3,1)$
- $[J_i, J_j] = i\epsilon_{ijk}J_k \implies$ forms a closed Algebra, but together J_i and K_i don't form a closed algebra.
- we define $N_i = J_i + iK_i$ and $\bar{N}_i = J_i - iK_i$

1.1 Lie Groups

- Groups with elements as smooth function¹ of some finite set of parameters $\theta_a \in \mathbb{R}$
- $g \in G$ is written as

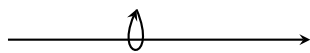
$$g = g(\theta_1, \theta_2, \dots, \theta_n) = e^{i\theta_a T^a} = e^{i\vec{\theta} \cdot \vec{T}} \quad (1.1)$$

where θ_a are continuous parameters

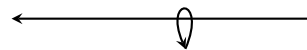
- In $U(1)$ group: e^{ieQ}
where e is charge and Q is generator

¹Continuous and differentiable

- In $SU(2)$ group: $e^{i\theta_a \tau_a}$
 τ_a are Pauli Matrices $\tau_a = \sigma_a/2$
- In $U(N)$ no of parameters = $2N^2$
 $U^\dagger U$ gives N^2 constraints, therefore N^2 elements are there
- In $SU(N)$ has $N^2 - 1$ generators, therefore in 3d 8 generators are there
- from particle physics perspective for each generator there is a mediator
- $U(1)$ describes photons



Spin anti-oriented
along momentum
Left handed



Spin oriented
along momentum
Right handed

- Parity group $\mathbb{Z}_2$

$\cdot$	e	a
e	e	a
a	a	e

- $\mathbb{Z}_n$ roots of $z^n - 1$